

Quantum Information - Formula Sheet

1 Lecture 1

Fundamental Thermodynamics & Equations of State

- **Ideal Gas Law:** $PV = Nk_B T = nRT$
- **First Law (Finite):** $\Delta U = \Delta Q - \Delta W$
- **First Law (Infinitesimal):** $dU = \delta Q - \delta W$
- **Work:** $dW = PdV$
- **Heat Capacity:** $\Delta Q = C\Delta T$

Heat Engines & Second Law

- **Work in Cycle:** $W = Q_2 - Q_1$
- **Engine Efficiency:** $\eta = \frac{W}{Q_2} = 1 - \frac{Q_1}{Q_2}$
- **Clausius Theorem:** $\oint \frac{dQ}{T} \leq 0$

Macroscopic Entropy

- **Entropy Change (Reversible):** $\Delta S = \int_{rev} \frac{dQ}{T}$
- **Third Law:** $S = 0$ at $T = 0$

Statistical Mechanics

- **Multiplicity:** $t(\{n_j\}) = \frac{N!}{\prod_j n_j!} \prod_j g_j^{n_j}$
- **Stirling's Approximation:** $\ln n! \approx n \ln n - n$

Boltzmann Distribution

- **Partition Function:** $Z = \sum_j \exp(\beta \epsilon_j)$
- **Occupation Numbers:** $n_j = \frac{N}{Z} \exp(\beta \epsilon_j)$
- **Lagrange Multiplier:** $\beta = -\frac{1}{k_B T}$

Microscopic Entropy

- **Boltzmann Entropy:** $S = k_B \ln \Omega$
- **Gibbs Entropy:** $S = -k_B \sum_{j=1}^m P_j \ln P_j$
- **Fundamental Relation:** $dU = TdS - PdV$

Energy Decomposition

- **Energy Change:** $dU = \sum_j \epsilon_j dn_j + \sum_j n_j d\epsilon_j$
- **Heat Component:** $\sum_j \epsilon_j dn_j$ (redistribution of particles)
- **Work Component:** $\sum_j n_j d\epsilon_j$ (shifting energy levels)